

4.6.1 Triple Integrals in Cylindrical Coordinates Worksheet

1. Change the following points from rectangular coordinates to cylindrical coordinates.

a) $(-1, 1, 1)$

b) $(-2, 2\sqrt{3}, 3)$

c) $(2\sqrt{3}, 2, -1)$

d) $(4, -3, 2)$ (round your angle measurement to 2 decimal places)

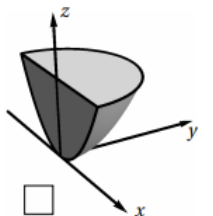
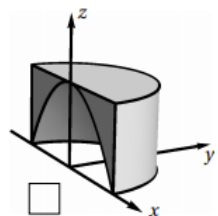
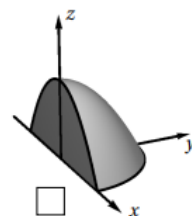
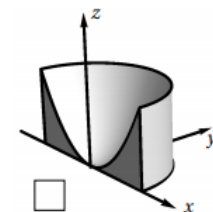
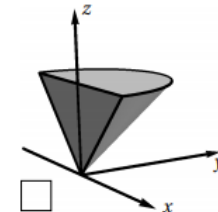
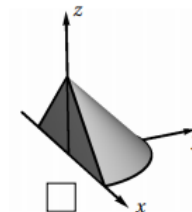
2. Describe in words the surface whose equation is given.

a) $\theta = \frac{\pi}{4}$

b) $r = 5$

3. For each of the given integrals, label the box below the picture of the corresponding region of integration in cylindrical coordinates.

a) $\int_0^\pi \int_0^1 \int_{1-r^2}^1 f(r, \theta, z) r \, dz \, dr \, d\theta$



b) $\int_0^\pi \int_0^1 \int_0^{1-z} f(r, \theta, z) r \, dr \, dz \, d\theta$

4. Sketch the solid whose volume is given by the integral and then evaluate the integral.

a) $\int_{-\pi/2}^{\pi/2} \int_0^2 \int_0^{r^2} r \, dz \, dr \, d\theta$

b) $\int_0^2 \int_0^{2\pi} \int_0^r r \, dz d\theta dr$

5. Evaluate $\iiint_E z \, dV$, where E is enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

6. Evaluate $\iiint_E (x + y + z) dV$, where E is the solid in the first octant that lies under the paraboloid $z = 4 - x^2 - y^2$.

7. Evaluate $\iiint_E x \, dV$, where E is enclosed by the planes $z = 0$ and $z = x + y + 5$ and by the cylinders $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$.

8. Find the volume of the solid that lies between the paraboloid $z = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 2$.

9. Evaluate the integral by changing to cylindrical coordinates.

a)
$$\int_{-\sqrt{3}}^{\sqrt{3}} \int_{-\sqrt{3-x^2}}^{\sqrt{3-x^2}} \int_1^{4-x^2-y^2} \frac{1}{z^2} dz dy dx$$

b)
$$\int_0^2 \int_{-\sqrt{4-y^2}}^0 \int_{\sqrt{x^2+y^2}}^2 xz dz dx dy$$

We see that the region of integration is given by:

$$\text{c) } \int_{-1}^1 \int_{-\sqrt{1-x^2}}^0 \int_{x^2+y^2}^1 \sqrt{x^2+y^2} \, dz dy dx$$

$$\text{d) } \int_{-4}^4 \int_0^{\sqrt{16-y^2}} \int_1^3 e^{-z} \, dz dx dy$$

10. Let E be the solid region bounded by the ellipsoid $2x^2 + 2y^2 + z^2 = 1$. Let $f(x, y, z) = \frac{1}{1 - x^2 - y^2 - z^2}$.

In the following parts we will try to $\iiint_E f(x, y, z) dV$ using cylindrical coordinates.

a) Set up the integral using cylindrical coordinates without bounds first and try to determine which order your differentials should be in to make your integration as easy as possible.

b) Fully setup up your integral with bounds.

c) Calculate the integral you found in (b). *Hint: After integration with respect to r consider using properties of logarithms to help simplify.*